

Course: IT202-008-S2025

Assignment: IT202 Module 4 SQL Challenge

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Status: Submitted | Worksheet Progress: 105.88%

Potential Grade: 10.90/10.00 (109.00%)

Received Grade: 0.00/10.00 (0.00%)

Grading Link: <https://learn.ethereallab.app/assignment/v3/IT202-008-S2025/it202-module-4-sql-challenge/grading/cle3>

Instructions

1. Ensure you read all instructions and objectives before starting.
2. Create a new branch from dev called M4-Homework
 1. `git checkout dev` (ensure proper starting branch)
 2. `git pull origin dev` (ensure history is up to date)
 3. `git checkout -b M4-Homework` (create and switch to branch)
3. Copy the template code from here: [GitHub Repository - M4 Homework](#)
 - Put all into an M4 folder inside your `public_html`
 - Expected files
 - ☐ `sql/000_create_table_todos.sql`
 - ☐ `sql/init_db.php`
 - ☐ `todos/create.php`
 - ☐ `todos/pending.php`
 - ☐ `todos/completed.php`
 - ☐ `nav.php`
 - Immediately record to history
 - ☐ `git add public_html`
 - ☐ `git commit -m "adding M4 HW baseline files"`
 - ☐ `git push origin M4-Homework`
 - ☐ Create a Pull Request from M4-Homework to dev and keep it open
4. Fill out the below worksheet
 - Each Problem requires the following as you work
 - ☐ Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
 - ☐ Update ucid in `nav.php` where noted
 - ☐ Code solution (add/commit periodically as needed) (refer to comments in the create, pending, completed files)
5. Once finished, click "Submit and Export"
6. Locally add the generated PDF to a folder of your choosing inside your repository folder and move it to Github
 1. `git add .`
 2. `git commit -m "adding PDF"`
 3. `git push origin M4-Homework`
 4. On Github merge the pull request from M4-Homework to dev
 5. On Github create a pull request from dev to prod and immediately merge. (This will trigger the prod deploy to make the heroku prod links work)

7. Upload the same PDF to Canvas

8. Sync Local

1. git checkout dev
2. git pull origin dev

Section #1: (1 pt.) Prep

Details:

Once the files are copied in, run your local PHP server and visit `/M4/sql/init_db.php` in the browser to trigger the table generation.

Task #1 (0.50 pts.) - Show the output of the init_db.php tool

Weight: 50%

Objective: Show the output of the `init_db.php` tool

Details:

The output should note either successfully executed or blocked (this is ok)

Image Prompt



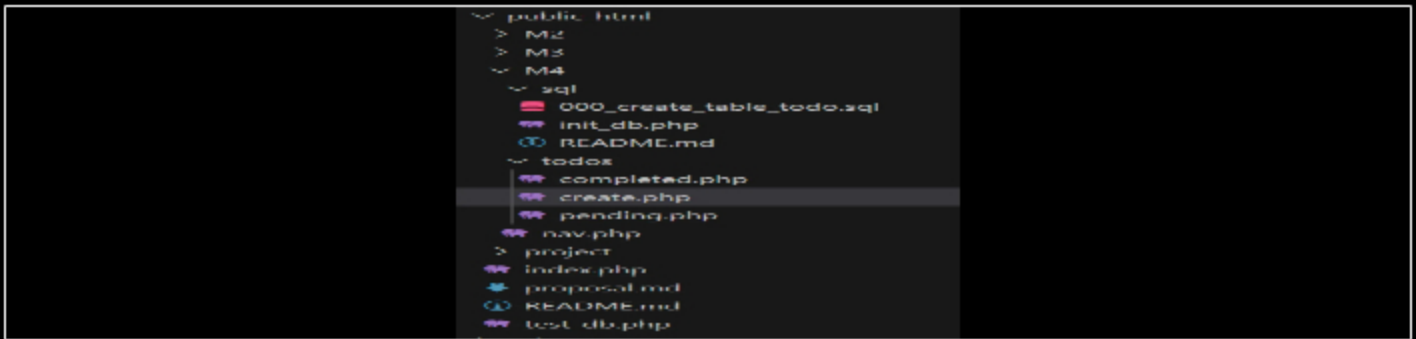
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Task #2 (0.50 pts.) - Show the project view of your editor to show a

Weight: 50%

Objective: Show the project view of your editor to show all files are present

Image Prompt



Showing projects in vscode



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Section #2: (3 pts.) Create Page

Task #1 (0.75 pts.) - HTML Form Challenge

Combo Task:

Weight: 25%

Objective: HTML Form Challenge

Item:#1

Weight: 50%

Details:

Show the code snippet related to the completed HTML form

Image Prompt



```

        design the form with proper labels and input fields with the correct types based on the SQL table.
        Wrap each label/input pair in a div tag.
        For assigned ensure the default value is "self" -->
<div>
  <label for="task">Task</label>
  <input type="text" id="task" name="task" required maxlength="200" />
</div>
<div>
  <label for="due">Due Date</label>
  <input type="date" id="due" name="due" required />
</div>
<div>
  <label for="assigned">Assigned</label>
  <input type="text" id="assigned" name="assigned" value="self" maxlength="50" />
</div>
<input type="button" value="Create Task" />
</div>
</form>
</body>
</html>

```

Showing the form



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Item:#3

Weight: 50%

Details:

Items

- Note the field names and why the specific types were chosen
- Also note how you made "self" the default for the assigned input

≡ Text Prompt

Your Response:

The names and the types I chose was because it needed to match the table structure. This is also seen when I assigned value = 'self'.



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Task #2 (0.75 pts.) - Form Submission PHP Validations

Combo Task:

Weight: 25%

Objective: Form Submission PHP Validations

Item:#1

Weight: 50%

Details:

- Validate the incoming data for correct format based on the SQL table definition.
- When not valid, provide a user-friendly message of what specifically was wrong and set `$is_valid` to false
- Assigned should check for "self" if a valid format/value isn't provided
- Include screenshot(s) from Heroku Dev demonstrating each validation (ensure URL is visible)

Image Prompt

```
// validate task
if (empty($task)) {
    echo "Task cannot be empty.<br>";
    $is_valid = false;
} elseif (strlen($task) > 255) { // Assuming task is a VARCHAR(255) in the database
    echo "Task must be 255 characters or fewer.<br>";
    $is_valid = false;
}

// validate due date
if (empty($due)) {
    echo "Due date cannot be empty.<br>";
    $is_valid = false;
} elseif (!preg_match('/^Y\d{4}-M\d{2}-D\d{2}$/', $due)) { // Validate MySQL date format (YYYY-MM-DD)
    echo "Due date must be in the format YYYY-MM-DD.<br>";
    $is_valid = false;
} elseif (strtotime($due) === false) { // Validate that the date is valid
    echo "Due date is not a valid date.<br>";
    $is_valid = false;
}

// validate assigned
if (empty($assigned)) {
    echo "Assigned cannot be empty. Defaulting to 'self'.<br>";
    $assigned = "self"; // Default to "self" if empty
} elseif (strlen($assigned) > 40) { // Assuming assigned is a VARCHAR(40) in the database
    echo "Assigned must be 40 characters or fewer. Defaulting to 'self'.<br>";
    $assigned = "self"; // Default to "self" if too long
}
```

screenshot for validation



Showing validation prompt

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Item:#3

Weight: 50%

Details:

Briefly explain your validation logic.

Text Prompt

Your Response:

Validation makes sure the task is not empty, and that the insertion is valid. For example, `$is_valid` is set to false and its used to check against invalid insertions.



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Task #3 (0.75 pts.) - Show the code snippet related to the INSERT c

Combo Task:

Weight: 25%

Objective: Show the code snippet related to the INSERT query

Item:#1

Weight: 50%

Details:

Design a query to insert the incoming data to the proper columns. Ensure valid and proper PDO named placeholders are used. <https://phpdelusions.net/pdo>

≡ Image Prompt

```
<?php
if ($is_data) {
    //
    Design a query to insert the incoming data to the proper columns.
    Ensure valid and proper PDO named placeholders are used.
    https://phpdelusions.net/pdo
    //
    $query = "INSERT INTO ps_todo (task, user, assigned) VALUES (:task, :user, :assigned)"; // edit this
    $params = [":task" => $task, ":user" => $user, ":assigned" => $assigned]; // apply the proper pdo placeholders to va
    $stmt = $pdo->prepare($query);
    $stmt->execute($params);
    if ($stmt) {
        echo "Inserted new todo with id " . $pdo->lastInsertId();
    } else {
        echo "Failed to insert";
    }
}
```

Showing insert query



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Item:#3

Weight: 50%

Details:

Briefly describe how the query works in plain words.

≡ Text Prompt

Your Response:

The insert query uses data objects to insert data into the table. This allows it to map and track the insertions and avoid overlaps/errors.



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Task #4 (+ 0.94 pts.) - Extra Credit: Handle Unique Constraint Errors

Combo Task:

Weight: 31.25%

Objective: *Extra Credit: Handle Unique Constraint Errors*

Item:#1

Details:

Show the code related to handling the unique constraint errors and the user-friendly message

≡ Image Prompt

```
> catch (PDOException $e) {  
    // error handle  
    // check if the exception was related to a unique constraint  
    // provide an appropriate user friendly message for this scenario  
    // otherwise show the default message below  
    if ((string)$e->getCode() === '23000') { //this is a universal unique constraint error exception code  
        echo "This already exists. Please provide a unique input.";  
    }  
    else {  
        echo "Error occurred during insertion; please check the logs in terminal";  
        error_log("Insert Error: " . var_export($e, true)); // shows in the terminal  
    }  
} -- 8/1-8/2 catch (PDOException $e)  
} else {  
    error_log("Creation input wasn't valid");  
}
```

Showing code for extra credit



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Item:#3

Details:

Briefly explain how your code solves the challenge.

≡ Text Prompt

Your Response:

The code works by checking if the error code is '23000', which in PHP tells us it is a unique constraint violation. If that error is present we then tell the user about it.



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Task #5 (0.75 pts.) - Create Page URLs

Weight: 25%

Objective: Create Page URLs

Details:

- Direct link to the file in the homework related branch from Github (should end in .php)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

🔗 Url Prompt

URL #1

<https://cle3-it202-008-prod-58562394d633.herokuapp.com/M4/todos/create.php>



URL

<https://cle3-it202-008-prod-58562394d633.herokuapp.com/M4/todos/create.php>



URL #2

https://github.com/carledouard-dev/cle3-IT202-008-M4-Homework/public_html/M4/todos/create.php



URL

https://github.com/carledouard-dev/cle3-IT202-008-M4-Homework/public_html/M4/todos/create.php



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Section #3: (3 pts.) Pending Page

Task #1 (1 pt.) - Fetching ToDos

Combo Task:

Weight: 33.33%

Objective: Fetching ToDos

Item:#1

Weight: 50%

Details:

- Refer to the HTML table below and build a query that'll select the columns in the same order as the table from the Todo table
- Cross-reference the HTML table columns with what'd most plausibly match the SQL table aside from the notes below
- For the Status part, you'll need to calculate the "days_offset" from the due date, ensure the virtual column matches "days_offset"
- For Actions, this isn't part of the query and there's nothing special to select for it
- Filter the results where the todo item is NOT completed and order the results by those due the soonest
- No limit is required
- Include screenshot(s) from Heroku Dev demonstrating 5 different records (ensure URL is visible)

Image Prompt

```
sqlquery "SELECT id, task, due,
            DATEDIFF(due, CURDATE()) AS days_offset,
            assigned
        FROM MA_Todos
        WHERE is_complete = 0
        ORDER BY due ASC"; // edit this
$results = [];
try {
    $stmt = $db->prepare($query);
    $r = $stmt->execute();
    if ($r) {
        $results = $stmt->fetchAll();
    }
} catch (PDOException $e) {
    echo "Error fetching pending todos: check the logs (terminal)";
    error_log("select error: " . var_export($e, true)); // shows in the terminal
}
</>
<html>
```

Showing fetch code

Pending ToDos					
ID	Task	Due Date	Status	Assigned	Actions

Showing correct output



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Item:#3

Weight: 50%

Details:

Briefly describe how the query works in plain words.

≡ Text Prompt

Your Response:

Query fetches incomplete tasks by using `is_complete = 0` from the `M4_Todos` table. It calculates the days and differences from tasks.



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Task #2 (1 pt.) - Completing Todos

Combo Task:

Weight: 33.33%

Objective: *Completing Todos*

Item:#1

Weight: 50%

Details:

- Create a query that'll update the respective `ToDo` marking it complete and setting the date for the completed date field as today
- Ensure the "id" is utilized using proper PDO named placeholders so that only the one item is updated
- Add an extra clause to update only if the complete field of the record is not set

≡ Image Prompt

Showing how todos were completed



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Item:#3

Weight: 50%

Details:

Briefly describe how the query works in plain words.

Text Prompt

Your Response:

It works by marking a task as complete (setting `is_complete` to 1) and checking against that to see if the task is completed.

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Task #3 (1 pt.) - Pending Page URLs

Weight: 33.33%

Objective: *Pending Page URLs*

Details:

- Direct link to the file in the homework related branch from Github (should end in .php)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

URL #1

<https://cle3-it202-008-prod-58562394d633.herokuapp.com/M4/todos/pending.php>



URL

<https://cle3-it202-008-prod-58562394d633.herokuapp.com/M4/todos/pending.php>



URL #2

https://github.com/carledouard-dev/cle3-IT202-008-M4-Homework/public_html/M4/todos/pending.php



URL

https://github.com/carledouard-dev/cle3-IT202-008-M4-Homework/public_html/M4/todos/pending.php



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Section #4: (2 pts.) Completed Page

Task #1 (1 pt.) - Fetching ToDos

Combo Task:

Weight: 50%

Objective: *Fetching ToDos*

Item:#1

Weight: 50%

Details:

- Refer to the HTML table below and build a query that'll select the columns in the same order as the table from the Todo table
- Cross-reference the HTML table columns with what'd most plausibly match the SQL table aside from the notes below
- For the completed date you'll need to extract the date portion from the completed column
- For the Status part, you'll need to calculate the "days_offset" from the completed date, ensure the virtual column matches "days_offset"
- Filter the results where the todo item is completed and order the results by most recently completed and most recently due
- No limit is required
- Include screenshot(s) from Heroku Dev demonstrating 5 different records (ensure URL is visible)

Image Prompt

```
/* Refer to the H1M1 table below and build a query that'll select the columns in the same order as the H1M1
cross-reference the H1M1 table columns with what'd most plausibly match the SQL table aside from the data
for the completed date you'll need to extract the date portion from the completed column.
For the status party, you'll need to calculate the "days_offset" from the completed date; ensure the var is
either the results where the task item is completed and order the results by most recently completed and
No limit is required.
*/
$query = "SELECT id, task, due,
          DATE(completed) AS completed_date,
          DATETIME(completed, due) AS days_offset,
          assigned
FROM M4_Todos
WHERE is_complete = 1
ORDER BY completed DESC, due DESC"; // edit this
$results = [];
try {
    $stmt = $db->prepare($query);
    if ($stmt->execute()) {
        if ($stmt->rowCount() > 0) {
            $results = $stmt->fetchAll();
        }
    } catch (PDOException $e) {
        echo "Error fetching completed todos; check the logs (terminal)";
        error_log("Fatal Error: " . var_export($e, true)); // shows in the terminal
    }
}
```

fetch code

Completed Todos					
ID	Task	Due Date	Completed Date	Days Offset	Assigned
1	Check assignments	2022-02-11	2022-02-11	Completed 0 days	Self
2	SQL	2022-02-09	2022-02-11	Completed 2 days	Self
3	PHP	2022-02-09	2022-02-11	Completed 2 days	Self
4	SQL	2022-02-09	2022-02-11	Completed 2 days	Self
5	PHP	2022-02-09	2022-02-11	Completed 2 days	Self

Showing heroku output completed



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Item:#3

Weight: 50%

Details:

Briefly describe how the query works in plain words.

Text Prompt

Your Response:

The query fetches completed tasks (is_complete = 1) from the M4_Todos table. It uses that to calculate and display completions based their offset order.



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Weight: 50%

Objective: Completed Page URLs

Details:

- Direct link to the file in the homework related branch from Github (should end in .php)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

Url Prompt

URL #1

<https://cle3-it202-008-prod-58562394d633.herokuapp.com/M4/todos/completed.php>



URL

<https://cle3-it202-008-prod-58562394d633.herokuapp.com/M4/todos/completed.php>



URL #2

https://github.com/carledouard-dev/cle3-IT202-008-M4-Homework/public_html/M4/todos/completed.php



URL

https://github.com/carledouard-dev/cle3-IT202-008-M4-Homework/public_html/M4/todos/completed.php



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Section #5: (1 pt.) Misc

Task #1 (0.33 pts.) - Github Details

Combo Task:

Weight: 33.33%

Objective: Github Details

Item:#1

Weight: 60%

Details:

From the Commits tab of the Pull Request screenshot the commit history

Image Prompt

The screenshot shows a GitHub pull request interface. At the top, it says 'Added M4 baseline files #8'. Below this, there's a table with columns for 'File', 'Status', and 'Size'. The table lists several files: 'Added M4 baseline files', 'Added README', 'Added README', 'Added README', 'Added README', and 'Added README'. Each row has a status icon (a green circle with a checkmark) and a size (e.g., 1 KB, 1 KB, 1 KB, 1 KB, 1 KB). Below the table, there's a section for 'commits' with a yellow arrow pointing down.

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Item:#2

Weight: 40%

Details:

Include the link to the Pull Request (should end in /pull/#)

Url Prompt

URL #1   <https://github.com/carledouard-dev/cle3-IT202404/> <https://github.com/carledouard-dev>

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Task #2 (0.33 pts.) - WakaTime - Activity

Weight: 33.33%

Objective: *WakaTime - Activity*

Details:

- Visit the WakaTime.com Dashboard
- Click Projects and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary

Projects • cle3-IT202-008

6 hrs 54 mins over the [Last 7 Days](#) in cle3-IT202-008 under [all](#) branches.

Waketime time

[illegible]

Task #3 (0.33 pts.) - Reflection

Sub-Tasks:

Task #1 (0.33 pts.) - What did you learn?

Weight: 33.33%

Objective: *What did you learn?*

Details:

Briefly answer the question (at least a few decent sentences)

≡, Text Prompt

Your Response:

I learned how to use objects and make tables, I also learned more about php and the commands/shortcuts that make my life easier. I got better with the commits, waiting until the end to push it to my remote repository.



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Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Weight: 33.33%

Objective: *What was the easiest part of the assignment?*

Details:

Briefly answer the question (at least a few decent sentences)

≡ Text Prompt

Your Response:

Easiest part was honestly creating the files, because all I had to do was make the name and copy paste the code from the Professor's example.



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Task #3 (0.33 pts.) - What was the hardest part of the assignment?

Weight: 33.33%

Objective: *What was the hardest part of the assignment?*

Details:

Briefly answer the question (at least a few decent sentences)

≡ Text Prompt

Your Response:

The hardest part was just human error on my end. I forgot to match the table structure, and it just made things harder.



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